

Lecture 20: Path Independence and conservative fields

Example: $\vec{F} = \langle y, x \rangle$ $\int_C \vec{F} \cdot d\vec{x}$, $C = c_1 \cup c_2 \cup c_3$

evaluating section of int. dot. $0 \leq \theta \leq \pi/4$ Need $\int_C y dx + x dy$

1) x -axis from $(0,0)$ to $(1,0)$: $y=0, dy=0$ $\int_{c_1} y dx + x dy = \int_0^1 0 dx + 0 = 0$ (Geometrically, on x -axis, $\vec{F}$ is $\parallel \hat{i}$, $\vec{F} \cdot \hat{i} = 0$)

2) c_2 : portion of unit circle $x = \cos \theta$ $y = \sin \theta$ $0 \leq \theta \leq \pi/4$ $dx = -\sin \theta d\theta$
 $dy = \cos \theta d\theta$

$$\int_{c_2} y dx + x dy = \int_0^{\pi/4} \sin \theta \cdot (-\sin \theta d\theta) + \cos \theta \cdot \cos \theta d\theta$$

$$\int_0^{\pi/4} \underbrace{\cos^2 \theta - \sin^2 \theta}_{\cos(2\theta)} d\theta = \int_0^{\pi/4} \frac{1}{2} \sin(2\theta) d\theta = \frac{1}{2}$$

$\int_{c_3}$ y -axis from $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ to $(0,0)$

We could do: $x = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}t$ $y = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}t$ with $0 \leq t \leq 1$ Better: $x=t$ $y=t$ from 0 to $\frac{1}{\sqrt{2}}$ gives us $(-c_3)$ (c_3 backwards)

$$\int_{c_3} y dx + x dy = \int_{\frac{1}{\sqrt{2}}}^0 t dt + t dt = \int_0^{\frac{1}{\sqrt{2}}} 2t dt = \left[t^2 \right]_0^{\frac{1}{\sqrt{2}}} = \frac{1}{2}$$

$$\int_{c_3} y dx + x dy = \frac{1}{2} \text{ Total work} = \int_C = \int_{c_1} + \int_{c_2} + \int_{c_3} = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

Special case: say $\vec{F} = \nabla \phi$ (grad. of field) ($\phi(x,y)$ is potential) Then we can simplify evaluation of $\int_C \vec{F} \cdot d\vec{x}$

Fundamental Theorem of calculus for line integrals:



$$\int_C \nabla F \cdot d\vec{x} = F(P_1) - F(P_0)$$

$$\int_C \left(\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \right) = \int_C dF = F(P_1) - F(P_0)$$

Proof: $\int_C \nabla F \cdot d\vec{x} = \int_C \left(\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \right)$ $C: (x(t), y(t))$
 $x = x(t), dx = x'(t) dt$
 $y = y(t), dy = y'(t) dt$

$$\int_C \nabla F \cdot d\vec{x} = \int_a^b \left(\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} \right) dt = \int_a^b \frac{dF}{dt} dt$$

$$\int_a^b \frac{dF}{dt} dt = \left[F(x(t), y(t)) \right]_a^b = F(P_1) - F(P_0)$$

Example: $\vec{F} = \langle y, x \rangle = \nabla F, F(x,y) = xy$ So $\int_C \vec{F} \cdot d\vec{x} = F(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) - F(0,0) = \frac{1}{2} - 0 = \frac{1}{2}$

(Warning: Only applies if $\vec{F}$ is a gradient field)

Consequences of Fundamental Theorem:

* Path-Independence

$$\int_{c_1} \nabla F \cdot d\vec{x} = \int_{c_2} \nabla F \cdot d\vec{x}$$

* $\vec{F} \cdot \nabla F = \text{conservative}$ $C = \text{closed curve}$

$$\int_C \vec{F} \cdot d\vec{x} = 0 \quad \left(\int_C \nabla F \cdot d\vec{x} = \oint (\text{not path}) = \oint (\text{trivial path}) \right)$$

Remarks: $\vec{F} = \langle y, x \rangle$ unit circle $\int_C \vec{F} \cdot d\vec{x} = \int_0^{2\pi} dt = 2\pi \neq 0$ On unit circle, $\vec{F} \perp \hat{t}$ $\vec{F} \cdot \hat{t} = |\vec{F}|/1 = 1$ Then not conservative and not path-independent, it's not a gradient field

Physics: If force $\vec{F}$ is gradient of a potential then work of $\vec{F}$ = change in value of potential

(e.g. gravitational field vs gravitational potential)
electrical field vs electrical potential

Lecture 20 part 2

→ conservativeness means no energy can be extracted from the field "for free"

- Total energy is conserved

Equivalent properties:

(1) $\vec{F}$ is conservative: $\oint_C \vec{F} \cdot d\vec{x} = 0$ along all closed curves C

(2) $\int_C \vec{F} \cdot d\vec{x}$ is path-independent

(3) $\vec{F}$ is a gradient field $\vec{F} = \langle f_x, f_y \rangle$

If (3) then (1) and (2) are true

If (2) and (1) then it will be how we find the potential

(4) $Mdx + Ndy$ is an exact differential - if